

DISSERTATION THESIS

ON

Two Treatment Covariate-Adjusted Response-Adaptive Allocation Design for Poisson Responses



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● Abstract

A clinical trial is a research activity which involves administration of a test treatment to some experimental units in order to evaluate the treatment. Adaptive allocation designs in clinical trials aim to enhance both ethical and statistical efficiency by dynamically adjusting the probability of assigning subjects to treatments based on accumulating data. This study proposes a two-treatment adaptive allocation design for clinical trials where responses follow a Poisson distribution. A class of covariate-adjusted adaptive allocation procedures is developed for a general class of responses with an aim to satisfy relevant clinical requirements.

● Introduction

Clinical studies refer to the class of all scientific approaches used to evaluate medical disease preventions, diagnostic techniques, and treatments. It involves clinical investigations conducted by pharmaceutical companies during various stages of development of pharmaceutical entities intended for marketing approval. An experimental unit is usually referred to as a subject from a targeted population under study.

This project focuses on the critical area of adaptive allocation strategies within clinical trials, a field of immense importance for both scientific rigor and ethical conduct in medical research. Among the various methodologies, CARA strategies have gained considerable attention due to their potential to enhance patient benefit during the course of a trial. These methods dynamically adjust the allocation of subjects to different treatment arms based on accumulated data, with the goal of favouring the more effective treatment.

Under the framework of this project, responses under each treatment are modelled as following a Poisson distribution, with parameters potentially influenced by a non-stochastic covariate u . That is, the distribution of the response for a patient depends on their covariate value, which is known and fixed.

● Objective

- To find the better treatment out of the two given treatments, we will use adaptive design to maintain the ethical purpose i.e. better the treatment, the more number of allocations to that treatment. To make the study more realistic we will introduce covariates to measure the effectiveness of the treatment.
- To make a correct decision about the treatment we will check which treatment results in less number of recurrence of the disease. Based on this we will propose test statistics for a demonstration purpose to select the better treatment.
- To check the consistency of the proposed test procedures for testing the superiority of the particular treatment over the other.

• Implications

- **Ethical Considerations:** Adaptive designs allow early stopping of trials if treatments are ineffective, reducing patient exposure to risk and enabling access to better alternatives. Covariate-aware modeling ensures that patients receive allocations based on both treatment efficacy and individual characteristics.
- **Statistical Precision:** In some cases, an adaptive design can provide a greater chance to detect a true drug effect (i.e., greater statistical power) than a comparable non-adaptive design. It may also achieve the same power with a smaller expected sample size or shorter duration.
- **Adaptive Design:** Allocation probabilities evolve over time and across patient subgroups defined by u , improving trial efficiency and ethical balance.

• CARA Method

Hu and Rosenberger (2006) define a covariate-adjusted response-adaptive (CARA) randomization procedure as one in which randomization probabilities for a current patient depend on the history of previous patients' treatment assignments, responses, and covariates. It expands upon the basic response-adaptive framework by utilizing both historical response data and baseline patient characteristics.

1. Use historic data to predict the outcome of patient n for each treatment.
2. When the next patient n enters, measure their baseline covariate value x_n , and use x_n along with historic data to predict outcomes for both treatments.
3. Assign the patient to their estimated best treatment with a certain probability.
4. Observe the outcome of patient n and update the estimated outcome of the selected treatment.
5. Repeat steps 2 to 4 for subsequent patients: $n + 1, n + 2, \dots, N$.

• Quantifying Treatment Superiority

The formulation of R_A as:

$$R_A = P[X_A < X_B] + \frac{1}{2}P[X_A = X_B]$$

here we use smaller the response better the treatment, thus it provides a natural and interpretable measure of how much better one treatment is than the other.

- When $R_A > \frac{1}{2}$, treatment A is preferred.

• General theory of Poisson Distribution

Let there be two competing treatments in a clinical trial where the response obtained from the treatments is recorded as count data.

Here,

X_K denotes the number of recurrences of a chronic disease after allocating k treatments $K = A, B$.

- We can use **smaller the response better the treatment** procedure.

- Thus we can use

$$X_K \sim \text{Poisson}, \quad K = A, B.$$

To make the study more realistic, introduce a covariate (prognostic variable) U . Assume that higher values of the covariate reduce the effectiveness of a treatment.

Thus,

$$X_K | U = u \sim \text{Poi}(\lambda_K(u)), \quad \forall K = A, B \text{ independently,}$$

where

$$\lambda_K(u) = E[X_K | U = u] = \lambda_K \cdot \Psi(u, a),$$

and $\Psi(u, a)$ is a function of u involving a constant a (which may be known or unknown).

Since lower response $\Leftrightarrow$ better performance.

Define the treatment allocation indicator:

$$\delta = \begin{cases} 1 & \text{if treatment A is applied} \\ 0 & \text{if treatment B is applied} \end{cases}$$

Define, in our case probability negatively affects the data. So, smaller the response, better the treatment

$$R_A(u) = P(\delta = 1 | u) = P(X_A < X_B | u) + \frac{1}{2}P(X_A = X_B | u),$$

$$R_B(u) = P(\delta = 0 | u) = P(X_A > X_B | u) + \frac{1}{2}P(X_A = X_B | u).$$

Note:

(i)

$$R_A(u) + R_B(u) = 1$$

since

$$P(X_A > X_B | u) + P(X_A < X_B | u) + P(X_A = X_B | u) = 1.$$

(ii) If X_A and X_B are identically distributed, then

$$P(X_A > X_B | u) = P(X_A < X_B | u)$$

$$\Rightarrow R_A(u) = R_B(u) = \frac{1}{2}, \quad \forall u > 0.$$

Proposed probability based allocation design and its characteristics

Let

$$R_A = P[X_A < X_B] + \frac{1}{2}P[X_A = X_B]$$

where smaller the response, better the treatment.

Hence, if $\text{Trt } A \cong \text{Trt } B$, then

$$P[X_A < X_B] = P[X_A > X_B]$$

(as then X_A & X_B are identically distributed).

Hence $R_A = R_B$, where $R_A + R_B = 1$

$$\Rightarrow R_A = \frac{1}{2} = R_B \text{ (i.e., the allocation rule is balanced)}$$

On the other hand, if $\text{Trt } A \succ \text{Trt } B$, then $\{X_A < X_B\}$ will be more likely than $\{X_A > X_B\}$.

$$\Rightarrow P[X_A < X_B] > P[X_A > X_B]$$

$\Rightarrow R_A > \frac{1}{2} > R_B$ i.e., probability based the allocation rule is skewed to the better treatment

Expanding this Poisson PMF yields —

$$\begin{aligned} R_A = & \sum_{r=0}^{\infty} \sum_{s=r+1}^{\infty} \frac{e^{-\lambda_A a^{-u}} (\lambda_A a^{-u})^r}{r!} \cdot \frac{e^{-\lambda_B a^{-u}} (\lambda_B a^{-u})^s}{s!} \\ & + \sum_{r=0}^{\infty} \frac{e^{-\lambda_A a^{-u}} (\lambda_A a^{-u})^r}{r!} \cdot \frac{e^{-\lambda_B a^{-u}} (\lambda_B a^{-u})^r}{r!} \end{aligned}$$

(This does not have a closed form.)

If λ_A and λ_B are known, $R_A(u)$ can be computed numerically or via simulation. However, in practice, λ_A and λ_B are unknown.

$\Rightarrow$ We need to estimate λ_A and λ_B from sample data.

• Maximum Likelihood Estimation of λ_A and λ_B (assuming a known):

Since patients are assigned randomly (with probability $R_A(u)$ to treatment A and $R_B(u)$ to treatment B), the likelihood function is:

$$\begin{aligned} L(\lambda_A, \lambda_B \mid \mathcal{F}_m) &\propto \prod_{i=1}^m \{f_{X_A}(x_{Ai} \mid u_i)\}^{\delta_i} \{f_{X_B}(x_{Bi} \mid u_i)\}^{1-\delta_i} \\ &= \text{Const.} \prod_{i=1}^m \left\{ \frac{e^{-\lambda_A a^{-u_i}} (\lambda_A a^{-u_i})^{x_{Ai}}}{x_{Ai}!} \right\}^{\delta_i} \left\{ \frac{e^{-\lambda_B a^{-u_i}} (\lambda_B a^{-u_i})^{x_{Bi}}}{x_{Bi}!} \right\}^{1-\delta_i} \end{aligned}$$

Log-likelihood:

$$l(\lambda_A, \lambda_B \mid \mathcal{F}_m) = \ln L(\lambda_A, \lambda_B \mid \mathcal{F}_m)$$

MLEs are obtained by solving:

$$\begin{aligned} \frac{\partial l}{\partial \lambda_A} = 0 &\Rightarrow \hat{\lambda}_{A,m} = \frac{\sum_{i=1}^m \delta_i x_{Ai}}{\sum_{i=1}^m \delta_i a^{-u_i}} \\ \frac{\partial l}{\partial \lambda_B} = 0 &\Rightarrow \hat{\lambda}_{B,m} = \frac{\sum_{i=1}^m (1 - \delta_i) x_{Bi}}{\sum_{i=1}^m (1 - \delta_i) a^{-u_i}} \end{aligned}$$

• Allocation Rule with Heuristic Approach

Simulation Methodology

We outline a simulation study conducted to evaluate the adaptive allocation rule as

• Simulation Procedure

Each of the 1000 independent simulation runs involved a total of 100 patients, where we take 8 patients for initial phase. Henceforth it is an adaptive design .

Performance Metric and Replications

The primary performance metric was the proportion of patients allocated to Treatment A (denoted as PropA) within each simulation run.

Take M = number of replication/repetition of the entire process for simulation study. we take $M = 1000$. The entire simulation process was replicated 1000 times to assess the average performance and variability of the allocation rule.

- **Proportion of Allocation:**

$$\text{propA} = \frac{\sum_{i=1}^n \delta_i}{n}, \quad \text{propB} = 1 - \text{propA}.$$

- **Average Allocation Proportions:**

$$\text{mean}(\text{propA}) = \frac{1}{M} \sum_{j=1}^M \text{propA}_j, \quad \text{mean}(\text{propB}) = \frac{1}{M} \sum_{j=1}^M \text{propB}_j.$$

If prior belief (or earlier evidence) suggests that Treatment A is superior to Treatment B, then typically

$$\text{mean}(\text{propA}) > \text{mean}(\text{propB}),$$

- **Variability Measure:**

The standard deviation of the allocation proportion for Treatment A is given by

$$\text{SD}(\text{propA}),$$

which measures the stability of the allocation rule. A smaller value indicates more consistent allocation performance.

- **Mean Squared Error (MSE) of Estimator propA:**

$$\text{MSE}(\text{propA}) = \frac{1}{M} \sum_{j=1}^M (\text{propA}_{n,j} - R_A)^2,$$

where $\text{propA}_{n,j} = \frac{\text{countA}}{n}$

Simulation Result

The performance of the adaptive allocation rule under different initial true parameter settings for the Poisson Distribution is summarized in Table 1 and Table 2 and Table 3 across the 1000 simulation replications

- For a fixed value of λ_A and the calculation is done by various values of λ_B

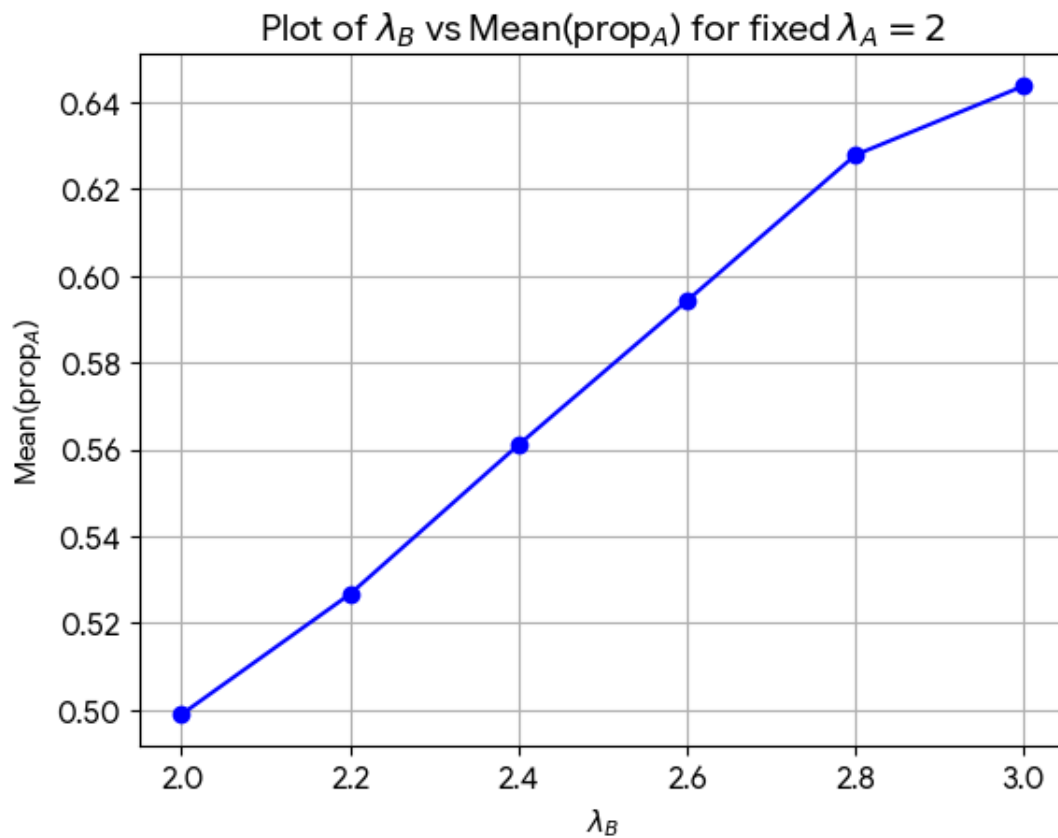


Figure 1: Mean allocation proportion vs λ_B for $\lambda_A = 2$

Table 1: $\lambda_A = 2$

λ_A	λ_B	Mean(prop _A)	SE	Bias	MSE
2	2.0	0.49897	0.1365272	-0.00363	0.01865286
2	2.2	0.52667	0.1315930	-0.00603	0.01735309
2	2.4	0.56103	0.1280837	-0.00747	0.01646125
2	2.6	0.59422	0.1297252	-0.00688	0.01687595
2	2.8	0.62780	0.1234025	-0.02460	0.01583335
2	3.0	0.64373	0.1185776	-0.02187	0.01453894

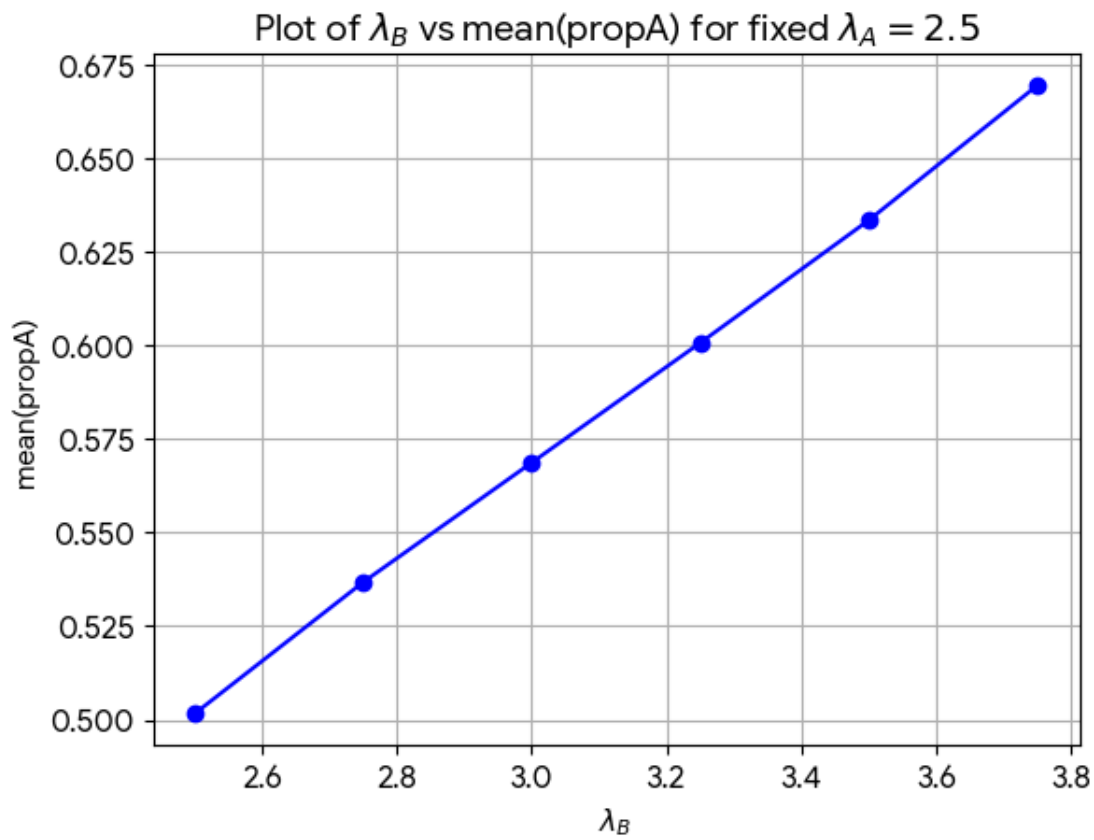


Figure 2: Mean allocation proportion vs λ_B for $\lambda_A = 2.5$

Table 2: $\lambda_A = 2.5$

λ_A	λ_B	Mean(p_A)	SE	Bias	MSE
2.5	2.5	0.50164	0.1341406	0.00824	0.01806160
2.5	2.75	0.53674	0.1368879	0.00324	0.01874881
2.5	3.0	0.56866	0.1282554	-0.01474	0.01666672
2.5	3.25	0.60063	0.1279706	-0.03407	0.01753724
2.5	3.5	0.63337	0.1219697	-0.03123	0.01585193
2.5	3.75	0.66935	0.1185206	-0.02385	0.01461595

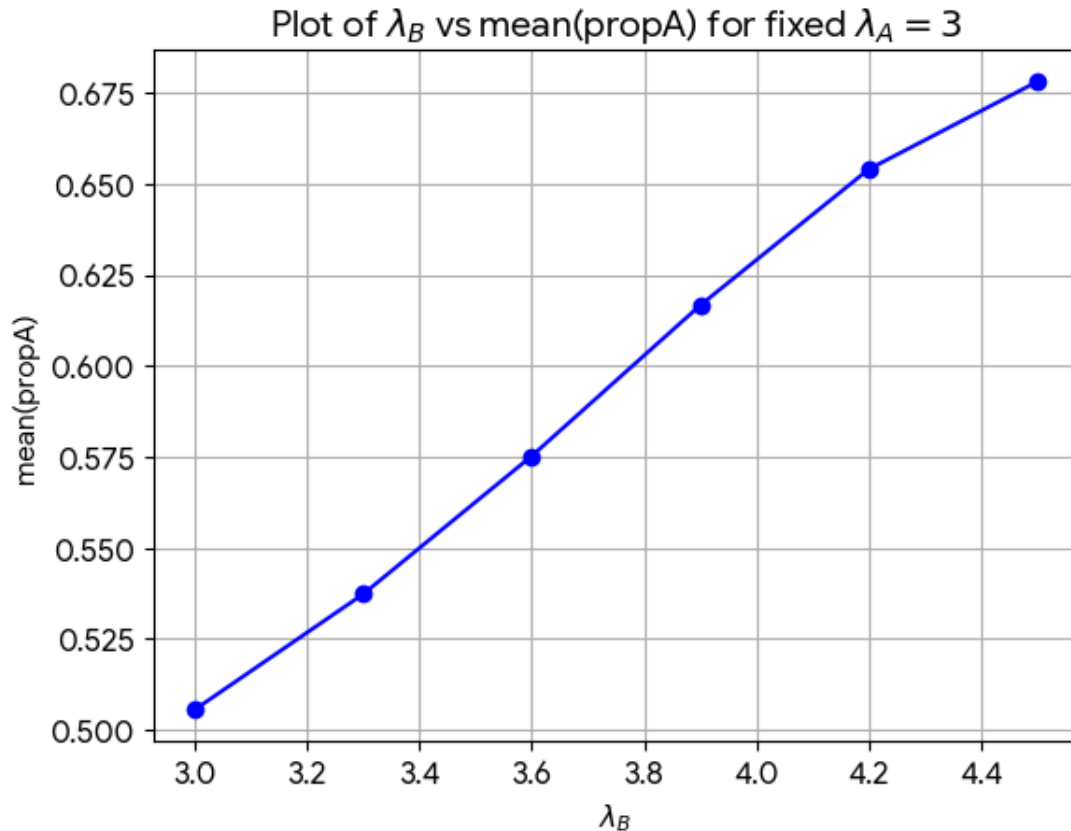


Figure 3: Mean allocation proportion vs λ_B for $\lambda_A = 3$

Table 3: $\lambda_A = 3$

λ_A	λ_B	Mean(p_A)	SE	Bias	MSE
3	3.0	0.50554	0.1315713	0.00214	0.01731560
3	3.3	0.53729	0.1336556	-0.02831	0.01866528
3	3.6	0.57522	0.1269656	-0.01068	0.01623433
3	3.9	0.61661	0.1274592	-0.01009	0.01634766
3	4.2	0.65399	0.1188070	-0.01521	0.01434644
3	4.5	0.67805	0.1221762	-0.02155	0.01539143

• Remark

• **ETHICS** - From all figures we can clearly see that when $\lambda_A = \lambda_B$, the mean is almost 0.5 for all three figures and as the value of λ_B increases by margin of 0.2, 0.25 and 0.3, the mean allocation proportion curve increases accordingly. As λ_B increases with respect to λ_A , that is treatment A becomes relatively more and more superior and the proposed allocation design assigns higher proportion of patients to superior treatment A. Therefore the ethics is maintained here, hence the proposed allocation design is established as an ethical allocation design.

• **SE** - generally mean is not related to std error but here as we are using poisson data which has mean and variance equal as λ , therefore as λ_B increases from λ_A , λ_A generally becomes smaller and therefore standard error (SE) of propA gradually decreases.

• Hypothesis Testing for Treatment Comparison

Suppose we want to test

H_0 : Treatment A and Treatment B are equivalent vs H_1 : Treatment A is better than Treatment B.

That is,

$$H_0 : \lambda_A(u) = \lambda_B(u) \quad \text{vs} \quad H_1 : \lambda_A(u) \leq \lambda_B(u),$$

with strict inequality for at least one u . In other words,

$$H_0 : \lambda_A = \lambda_B, \quad H_1 : \lambda_A < \lambda_B.$$

Asymptotic Distribution

By Martingale CLT,

$$\sqrt{m} \begin{pmatrix} \hat{\lambda}_{A,m} - \lambda_A \\ \hat{\lambda}_{B,m} - \lambda_B \end{pmatrix} \xrightarrow{\mathcal{D}} N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{\lambda_A}{\varphi_A} & 0 \\ 0 & \frac{\lambda_B}{\varphi_B} \end{pmatrix} \right).$$

Hence,

$$\sqrt{m} [(\hat{\lambda}_{A,m} - \hat{\lambda}_{B,m}) - (\lambda_A - \lambda_B)] \xrightarrow{\mathcal{D}} N \left(0, \frac{\lambda_A}{\varphi_A} + \frac{\lambda_B}{\varphi_B} \right).$$

Under Null Hypothesis

Under H_0 , $\lambda_A = \lambda_B = \lambda$.

The log-likelihood becomes:

$$l_{H_0}(\lambda \mid \mathcal{F}_n) = \text{const} - \lambda \sum_{i=1}^n a^{-U_i} + (\ln \lambda) \left[\sum_{i=1}^n \delta_i X_{Ai} + \sum_{i=1}^n (1 - \delta_i) X_{Bi} \right].$$

$$\hat{\lambda}_n = \frac{\hat{\lambda}_A \sum_{i=1}^n \delta_i a^{-u_i} + \hat{\lambda}_B \sum_{i=1}^n (1 - \delta_i) a^{-u_i}}{\sum_{i=1}^n a^{-u_i}}$$

Also, under H_0 ,

$$R_A(u) = R_B(u) = \frac{1}{2}.$$

Thus,

$$\varphi_A = \varphi_B = \varphi_0 = \frac{1}{2} E(a^{-U}).$$

Estimate via simulation:

$$\hat{\varphi}_0 = \frac{1}{2} \cdot \frac{1}{m} \sum_{j=1}^m a^{-U_j}.$$

Test Statistic - 1

As $n \rightarrow \infty$,

$$\frac{\sqrt{n}(\hat{\lambda}_{A,n} - \hat{\lambda}_{B,n})}{\sqrt{\hat{\lambda}_n \cdot \frac{2}{\hat{\varphi}_0}}} \xrightarrow{\mathcal{D}} N(0, 1).$$

Hence, the test statistic is:

$$T_1 = \frac{\sqrt{n}(\hat{\lambda}_{A,n} - \hat{\lambda}_{B,n})}{\sqrt{\frac{2\hat{\lambda}_n}{\hat{\varphi}_0}}}.$$

Another Test Statistic Formation

As $\lambda^{2/3}$ is a normalizing transformation.

Hence we propose another test statistic under consideration as –

Asymptotic part

$$\sqrt{m} \begin{pmatrix} \hat{\lambda}_{A,m}^{2/3} - \lambda_A^{2/3} \\ \hat{\lambda}_{B,m}^{2/3} - \lambda_B^{2/3} \end{pmatrix} \xrightarrow{\mathcal{D}} N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{4\lambda_A^{1/3}}{9\varphi_A} & 0 \\ 0 & \frac{4\lambda_B^{1/3}}{9\varphi_B} \end{pmatrix} \right).$$

Test Statistic - 2

As $n \rightarrow \infty$, under H_0

$$\frac{\sqrt{n}(\hat{\lambda}_{A,n}^{2/3} - \hat{\lambda}_{B,n}^{2/3})}{\frac{2}{3}\sqrt{\hat{\lambda}_n^{1/3} \cdot \frac{2}{\hat{\varphi}_0}}} \xrightarrow{\mathcal{D}} N(0, 1).$$

Hence the test statistic is :

$$T_2 = \frac{\sqrt{n}(\hat{\lambda}_{A,n}^{2/3} - \hat{\lambda}_{B,n}^{2/3})}{\frac{2}{3}\sqrt{\hat{\lambda}_n^{1/3} \cdot \frac{2}{\hat{\varphi}_0}}}$$

Decision Rule

Reject H_0 in favour of H_1 if:

$$T_n < C_\gamma,$$

where C_γ is the left-tailed critical value at level γ and is calculated by Monte Carlo simulation

Based on this critical value and the chosen level of significance, the power of the test was subsequently calculated.

Power

In this problem, the power is also calculated empirically. To compute the power of the test, another m ($= 1000$) fresh samples were generated by simulation. Based on these samples, the corresponding m ($= 1000$) test statistic values were calculated under the alternative hypothesis, i.e.,

$$\lambda_A < \lambda_B \quad (1)$$

Based on the previously determined critical value, the power of the test was computed by taking the proportion of m test statistic values (calculated under the alternative hypothesis) that lie under the critical value.

• Power Evaluation

The power of the test was computed empirically for different values of (λ_A, λ_B) such that $\lambda_A \leq \lambda_B$ using simulation over different functions under different distributions.

In this study, we evaluated and compared the power of the test across two different functional forms of u under two distinct distributional settings:

1. **Continuous Case:** The distribution was assumed to be normal with a mean of 10 and a standard deviation of 3, i.e., $N(10, 3^2)$. Under this setup, the power was compared between the two functional forms of u with 1000 simulation.

Comparison: $\psi(u) = a^{-u}$ vs $\psi(u) = \frac{e^u}{1 + e^u}$

Fixed Value: $\lambda_A = 2$

SL. No.	λ_B	Power using a^{-u}	Power using $\frac{e^u}{1 + e^u}$
1	2.4	0.115	0.096
2	2.8	0.222	0.135
3	3.2	0.370	0.190
4	3.6	0.504	0.243
5	4.0	0.675	0.317
6	4.4	0.820	0.380
7	4.8	0.896	0.469
8	5.2	0.931	0.481
9	5.6	0.967	0.566
10	6.0	0.981	0.653

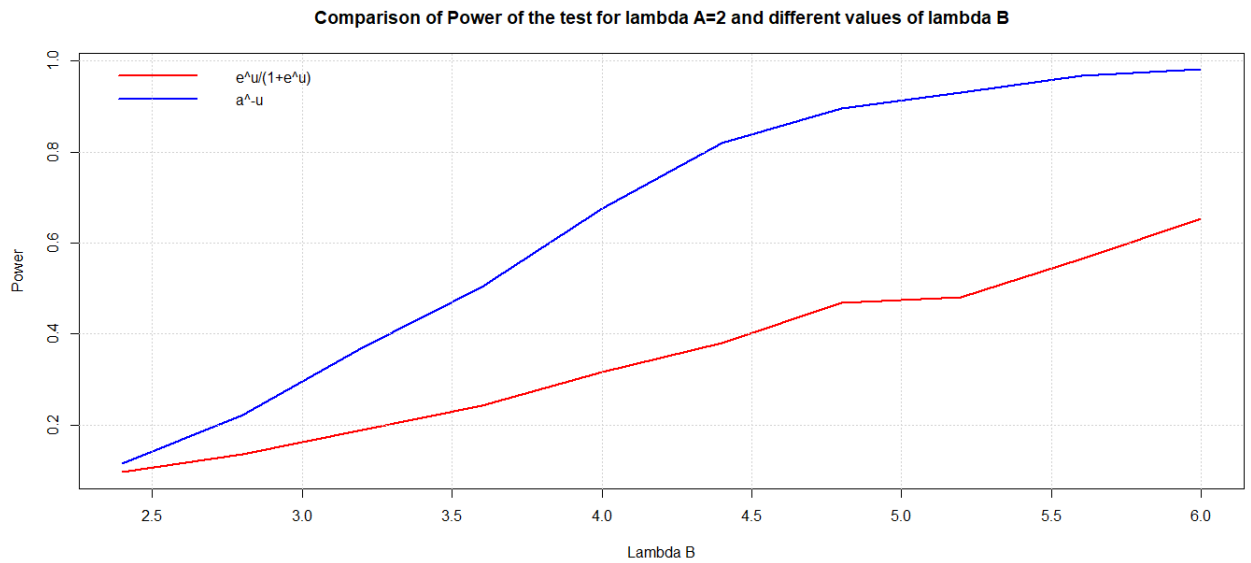


Figure 1: RED = $e^u/(1 + e^u)$ & BLUE = a^{-u}

2. **Discrete Case:** A discrete distribution was considered where the variable takes the values 0, 1, and 2 with probabilities 0.25, 0.50, and 0.25, respectively. Similarly, the power comparison was conducted between the two different functional forms of u under this discrete framework.

Comparison: $\psi(u) = a^{-u}$ vs $\psi(u) = \frac{u+1}{u+2}$

Fixed Value: $\lambda_A = 2$

SL. No.	λ_B	Power using a^{-u}	Power using $\frac{u+1}{u+2}$
1	2.4	0.081	0.095
2	2.8	0.124	0.129
3	3.2	0.185	0.171
4	3.6	0.242	0.226
5	4.0	0.328	0.255
6	4.4	0.389	0.295
7	4.8	0.480	0.359
8	5.2	0.630	0.420
9	5.6	0.634	0.497
10	6.0	0.684	0.538

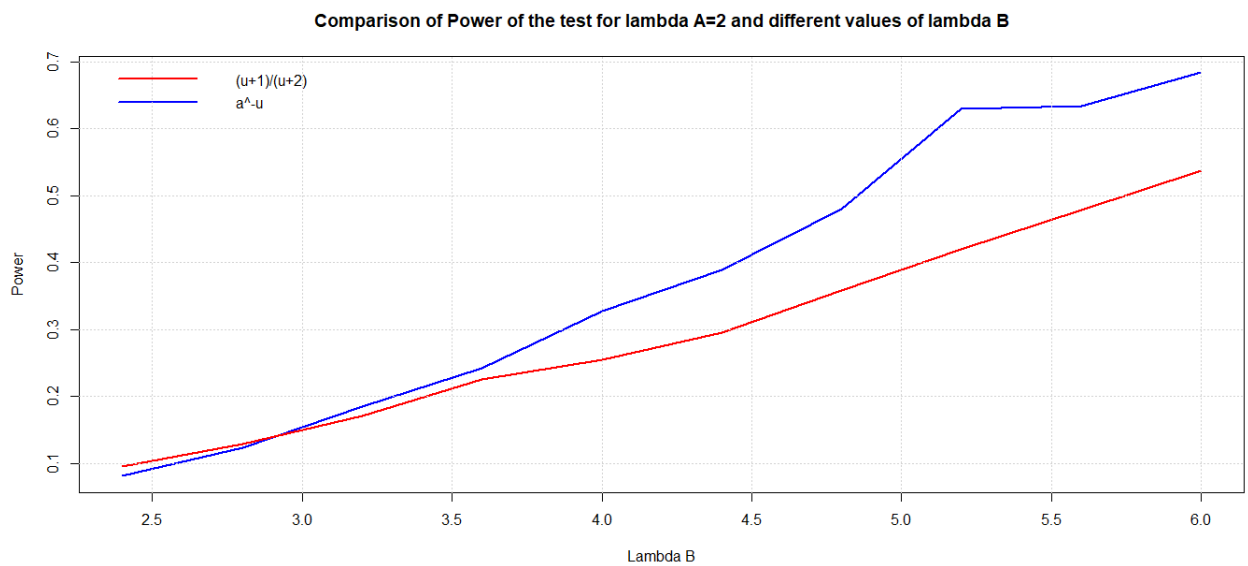


Figure 2: RED = $\frac{u+1}{u+2}$ & BLUE = a^{-u}

• Under two test statistics

3. **Continuous Case with two different test statistics for a^{-u} :** The distribution was assumed to be normal with a mean of 10 and a standard deviation of 3, i.e., $N(10, 3^2)$. Under this setup, the power was compared between the two test statistics for a^{-u} with 1000 simulation.

Here T_1 = Test statistic 1 & T_2 = Test statistic 2

i) For $\psi(u) = a^{-u}$ (Fixed $\lambda_A = 2$)

SL. No.	λ_B	Power for T_1	Power for T_2
1	2.4	0.115	0.131
2	2.8	0.222	0.235
3	3.2	0.370	0.398
4	3.6	0.504	0.556
5	4.0	0.675	0.693
6	4.4	0.820	0.828
7	4.8	0.896	0.895
8	5.2	0.931	0.947
9	5.6	0.967	0.976
10	6.0	0.981	0.986

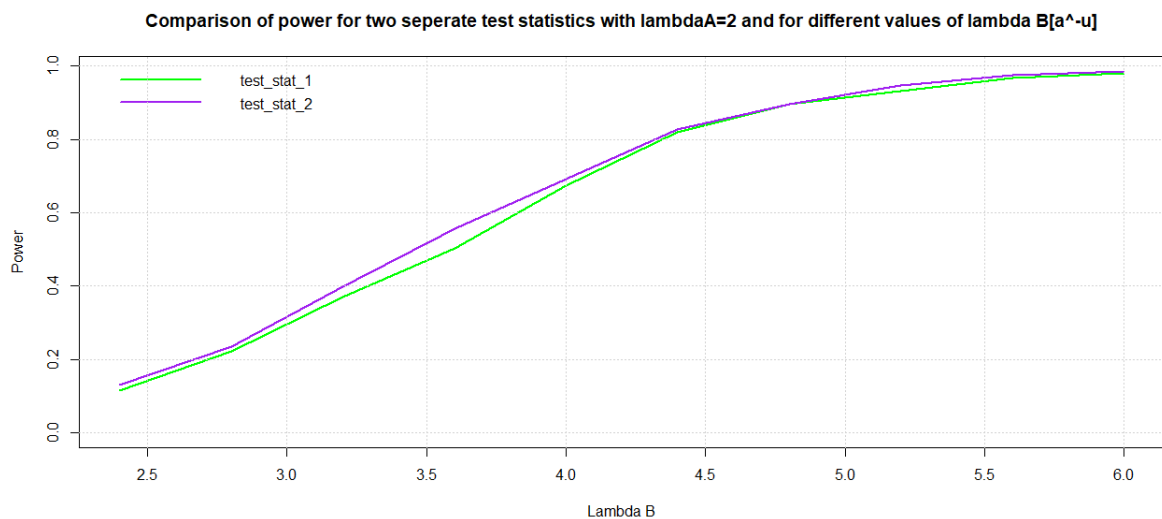


Figure 3: GREEN = TEST STATISTIC 1 & PURPLE = TEST STATISTIC 2

4. **Continuous Case with two different test statistics for $\frac{e^u}{1+e^u}$:** The distribution was assumed to be normal with a mean of 10 and a standard deviation of 3, i.e., $N(10, 3^2)$. Under this setup, the power was compared between the two test statistics for $\frac{e^u}{1+e^u}$ with 1000 simulation.

ii) For $\psi(u) = \frac{e^u}{1+e^u}$ (Fixed $\lambda_A = 2$)

SL. No.	λ_B	Power for T_1	Power for T_2
1	2.4	0.096	0.072
2	2.8	0.135	0.107
3	3.2	0.190	0.175
4	3.6	0.243	0.226
5	4.0	0.317	0.303
6	4.4	0.380	0.371
7	4.8	0.469	0.447
8	5.2	0.481	0.479
9	5.6	0.566	0.567
10	6.0	0.653	0.660

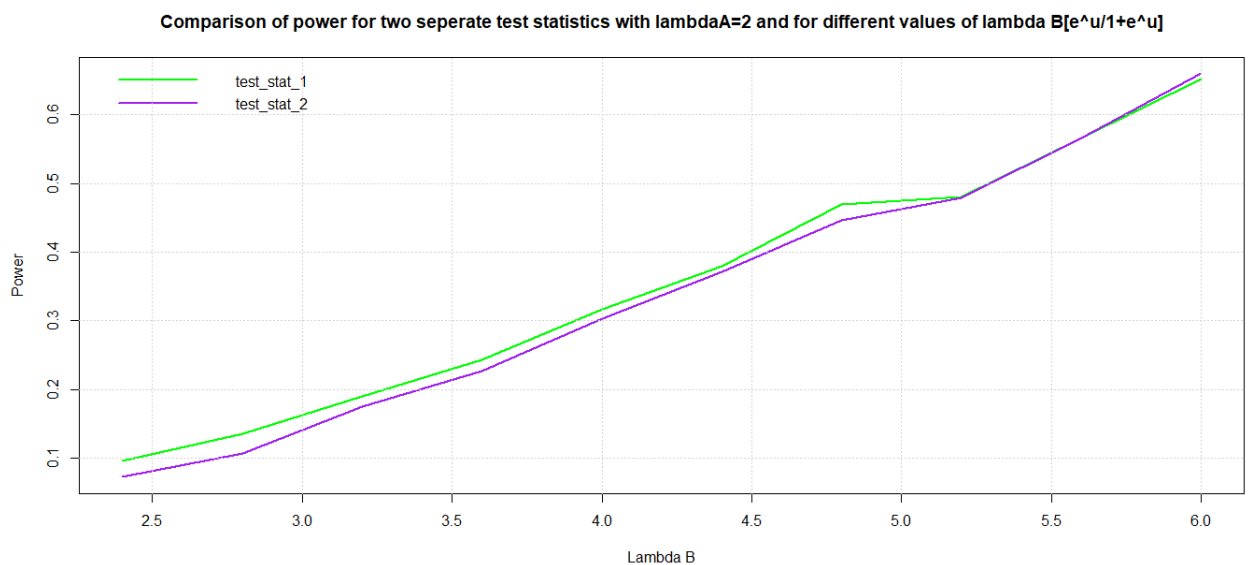


Figure 4: GREEN = TEST STATISTIC 1 & PURPLE = TEST STATISTIC 2

5. **Discrete Case with two different test statistics for a^{-u} :** A discrete distribution was considered where the variable takes the values 0, 1, and 2 with probabilities 0.25, 0.50, and 0.25, respectively. Similarly, the power comparison was conducted between the two different Statistics for a^{-u} under this discrete framework.

i) For $\psi(u) = a^{-u}$ (Fixed $\lambda_A = 2$)

SL. No.	λ_B	Power for T_1	Power for T_2
1	2.4	0.081	0.084
2	2.8	0.124	0.135
3	3.2	0.185	0.196
4	3.6	0.242	0.264
5	4.0	0.328	0.360
6	4.4	0.389	0.414
7	4.8	0.480	0.577
8	5.2	0.630	0.650
9	5.6	0.634	0.666
10	6.0	0.684	0.727

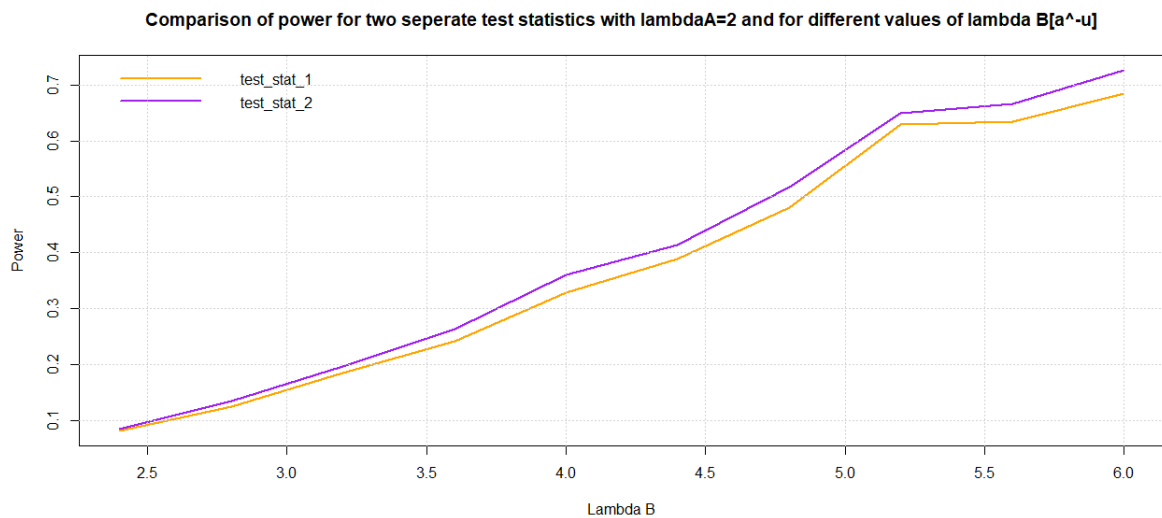


Figure 5: ORANGE = TEST STATISTIC 1 & PURPLE = TEST STATISTIC 2

6. **Discrete Case with two different test statistics for $(u + 1)/(u + 2)$** — A discrete distribution was considered where the variable takes the values 0, 1, and 2 with probabilities 0.25, 0.50, and 0.25, respectively. Similarly, the power comparison was conducted between the two different Statistics for $(u + 1)/(u + 2)$ under this discrete framework.

ii) For $\psi(u) = \frac{u + 1}{u + 2}$ (Fixed $\lambda_A = 2$)

SL. No.	λ_B	Power for T_1	Power for T_2
1	2.4	0.095	0.086
2	2.8	0.129	0.119
3	3.2	0.171	0.157
4	3.6	0.226	0.214
5	4.0	0.255	0.229
6	4.4	0.295	0.277
7	4.8	0.359	0.353
8	5.2	0.420	0.410
9	5.6	0.497	0.484
10	6.0	0.538	0.533

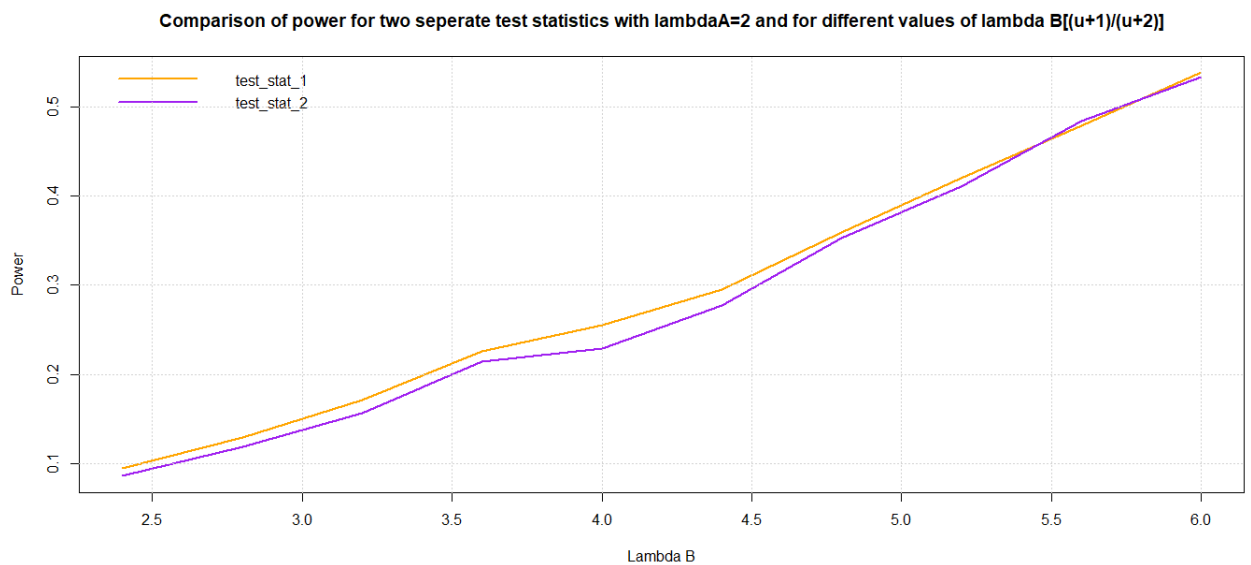


Figure 6: ORANGE = TEST STATISTIC 1 & PURPLE = TEST STATISTIC 2

Remark

Below are the structured analysis tables evaluating the power characteristics —

————— CHECKLIST —————

1. Functional Form Variations

Table 4: Same test statistic with different functional forms under continuous and discrete cases –

Distribution & Covariate	Functional Form	Decision & Perfrance
Continuous Case $u \sim N(10, 9)$	$e^u/(1 + e^u)$ vs. a^{-u}	Better: a^{-u} power curve for a^{-u} is better than $e^u/(1+e^u)$
Discrete Case $u \in \{0, 1, 2\}$ with w.p. $\{0.25, 0.50, 0.25\}$	$(u + 1)/(u + 2)$ vs. a^{-u}	Better: a^{-u} power curve for a^{-u} performs better than $(u + 1)/(u + 2)$

for both cases the functional form a^{-u} performs better.

2. Test Statistic Variations

Table 5: functional form same with different test statistic under continuous and discrete cases

Distribution & Covariate	Functional Form	Test Statistics	Decision & Performance
Continuous Case $u \sim N(10, 9)$	a^{-u}	1 & 2	Tie (Equivalence): Power curves are almost equal which implies they are comparable.
Continuous Case $u \sim N(10, 9)$	$\frac{e^u}{1 + e^u}$	1 & 2	Better: Test Statistic 1 Test Statistic 1 slightly performs better.
Discrete Case $u \in \{0, 1, 2\}$	a^{-u}	1 & 2	Better: Test Statistic 2 Test Statistic 2 performs slightly better.
Discrete Case $u \in \{0, 1, 2\}$	$(u + 1)/(u + 2)$	1 & 2	Better: Test Statistic 1 Test Statistic 1 has relatively better power curve .

Test Statistic 1 performs slightly better than Test Statistic 2 .

• Verification of Consistency of the test

Here are the figures and nature of the curve of consistency obtained by using different functions. These curves help us to understand the nature of the functions and their performances to make a conclusion. We have fixed the values of $\lambda_A = 2$ and $\lambda_B = 4$.

Here T_1 = Test statistic 1 & T_2 = Test statistic 2

Case 1 - Discrete Case:

$$\psi(u) = a^{-u}$$

SL. No.	Sample Size	Power for T_1	Power for T_2
1	100	0.337	0.361
2	150	0.431	0.465
3	200	0.493	0.525
4	250	0.552	0.592
5	300	0.581	0.631
6	350	0.602	0.636
7	400	0.613	0.641



Figure 7: Consistency discrete (a^{-u})

$$\psi(u) = \frac{u+1}{u+2}$$

SL. No.	Sample Size	Power for T_1	Power for T_2
1	100	0.280	0.273
2	150	0.361	0.365
3	200	0.399	0.415
4	250	0.438	0.459
5	300	0.475	0.494
6	350	0.499	0.511
7	400	0.526	0.554

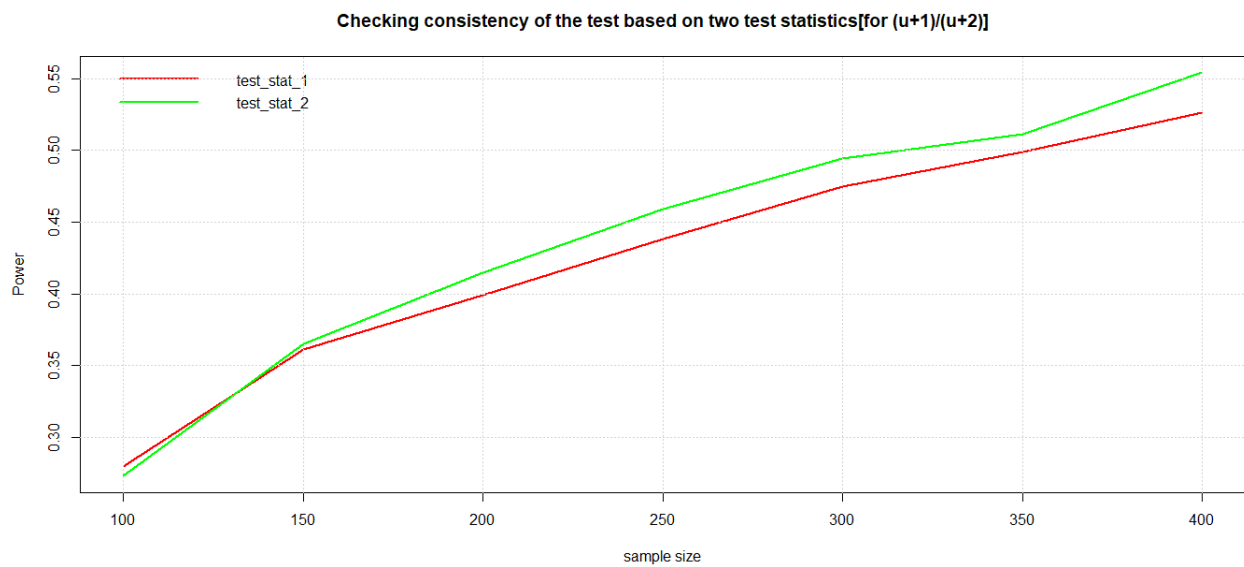


Figure 8: Consistency discrete $\left(\frac{u+1}{u+2}\right)$

Case 2 - Continuous Case:

$$\psi(u) = a^{-u}$$

SL. No.	Sample Size	Power for T_1	Power for T_2
1	100	0.690	0.710
2	150	0.780	0.796
3	200	0.838	0.809
4	250	0.864	0.865
5	300	0.867	0.891
6	350	0.864	0.883
7	400	0.887	0.890

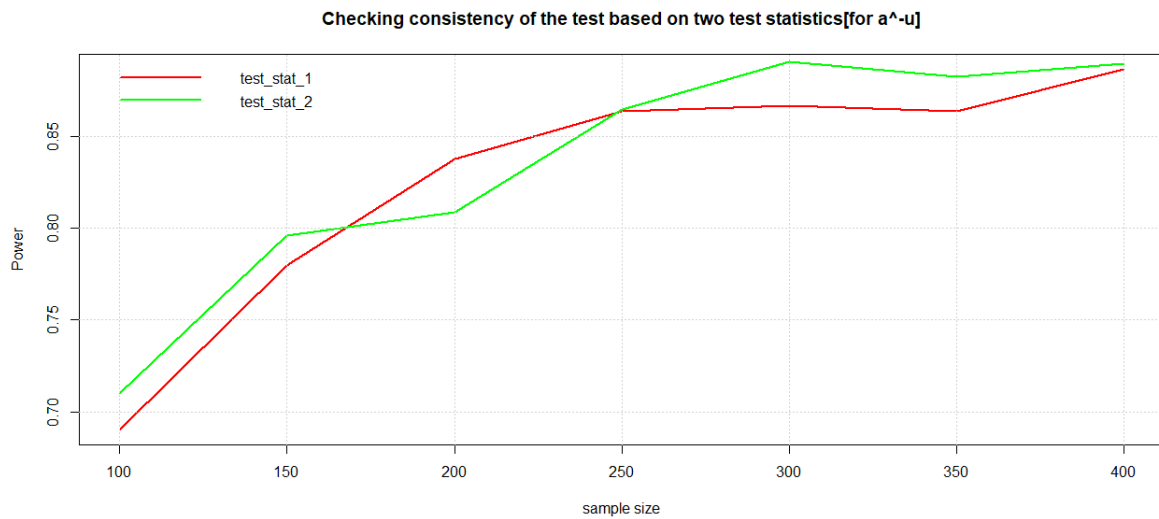


Figure 9: Consistency normal (a^{-u})

$$\psi(u) = \frac{e^u}{1 + e^u}$$

SL. No.	Sample Size	Power for T_1	Power for T_2
1	100	0.320	0.281
2	150	0.330	0.346
3	200	0.346	0.403
4	250	0.338	0.421
5	300	0.336	0.478
6	350	0.328	0.485
7	400	0.352	0.523

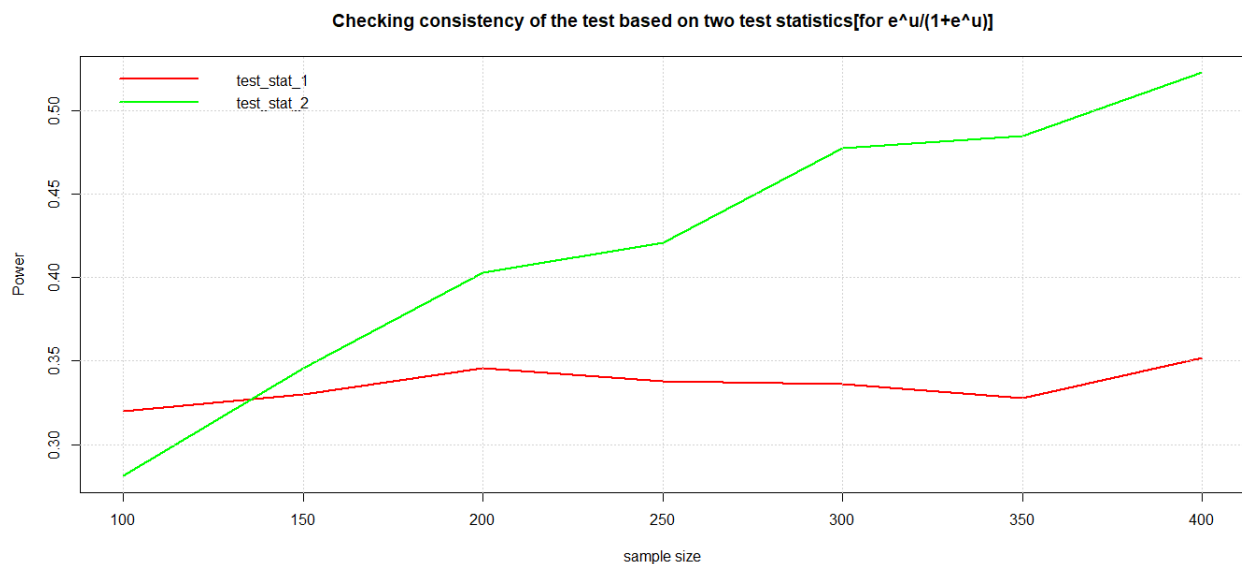


Figure 10: Consistency normal $\left(\frac{e^u}{1+e^u}\right)$

Remark:

- **Case 1(Discrete):** For both the functions, the power curve for test statistic 2 always lies above the power curve for test statistic 1. Thus test statistic 2 are more consistent than test statistic 1.
- **Case 2(Continuous):** The power graph for a^{-u} seems to be consistent for both the test statistics, while for $\frac{e^u}{1+e^u}$ test statistic 2 is more consistent as compared to the other.

Conclusion & Future Cases

Coming to the conclusion we can look at our design as follows—

- Ethics is maintained throughout the design which is a very important factor for any clinical trial, here we have studied for two treatments .
- We have considered the response of the count data which works extremely well for recurrent disease studies.
 - smaller the response = better the treatment
 - we have considered an important covariate for our study which is very important to maintain two aspects –

Ethical precision - maintained by mean allocation proportion curves .

Statistical precision -

- SE and MSE for estimator
- power curves of test procedure for testing.

Future Cases -

- We can also work with multi-treatment cases, i.e., with $K > 2$
- Recurrence of a disease with count data response is not the only parameter of interest.
- We can extend our idea towards discovering "**How long the disease stays.**" Basically it can be thought of as a sickness period .
- We can extend our idea towards discovering "**How long the patient stays healthy between recurrences of the disease**" Basically it can be thought of as a wellness period .

if we consider this, the study becomes complicated, but it becomes more realistic.

Appendix :

- The complete R simulation script that implements the 1000 replication covariate-adjusted response-adaptive (CARA) trial model for Poisson outcomes, along with its empirical power and consistency evaluations, is given below:

```
#project_sem4_PG
myfunction=function(N,M,lambda_A,lambda_B) #N denotes no. of patient
{
  countA=NULL
  propA=NULL
  T1=NULL      #for test statistic 1
  T2=NULL      #for test statistic 2
  for(z in 1:M)
  {
    X=NULL
    u=rnorm(N,mean=10,sd=3)
    u.trial=u[1:8]
    n=length(u.trial)
    a=0.9
    delta=NULL
    for(i in 1:n)
    {
      if(i<=n/2)
      {
        X[i]=rpois(n=1,lambda=lambda_A*a^(-u.trial[i]))
        delta[i]=1
      }
      else
      {
        X[i]=rpois(n=1,lambda=lambda_B*a^(-u.trial[i]))
        delta[i]=0
      }
    }
    Lam_nu.A=0
    Lam_de.A=0
    Lam_nu.B=0
    Lam_de.B=0
    for(i in 1:n)
    {
      if(delta[i]==1)
      {
        Lam_nu.A=Lam_nu.A+X[i]
        Lam_de.A=Lam_de.A+a^(-u[i])
      }
      else
      {
        Lam_nu.B=Lam_nu.B+X[i]
        Lam_de.B=Lam_de.B+a^(-u[i])
      }
    }
    lA.hat=Lam_nu.A/Lam_de.A
    lB.hat=Lam_nu.B/Lam_de.B
  }
  #-----function to define R_hat
  R.hat=function(m)
  {
    X_a=rpois(5000,lambda=lA.hat)
  }
}
```

```

X_b=rpois(5000,lambda=lB.hat)
p1=0
p2=0
for(j in 1:5000)
{
  if(X_a[j]<X_b[j])
    p1=p1+1
  else if(X_a[j]==X_b[j])
    p2=p2+1
  else
  {
    p1=p1
    p2=p2
  }
}
Ra.hat=(p1/5000)+0.5*(p2/5000)
return(Ra.hat)
}
for(i in (n+1):N)
{
  Ra.hat=R.hat(i-1)
  r=runif(1)
  if(r<=Ra.hat)
  {
    Xa=rpois(1,lambda=lA.hat*a^(-u[i]))
    X=c(X,Xa)
    delta[i]=1
    Lam_nu.A=Lam_nu.A+X[i]
    Lam_de.A=Lam_de.A+a^(-u[i])
    lA.hat=Lam_nu.A/Lam_de.A
  }
  else
  {
    Xb=rpois(1,lambda=lB.hat*a^(-u[i]))
    X=c(X,Xb)
    delta[i]=0
    Lam_nu.B=Lam_nu.B+X[i]
    Lam_de.B=Lam_de.B+a^(-u[i])
    lB.hat=Lam_nu.B/Lam_de.B
  }
}
countA[z]=sum(delta)
propA[z]=countA[z]/N
lambda_n.hat=( lA.hat*sum(delta*a^-u) + lB.hat*sum((1-delta)*a^-u) )/
sum(a^(-u))
Shi_n.hat=(0.5)*(exp( 10*(-log(a)) + (9*(-log(a))^2)/2) )
T1[z]=(sqrt(N)*(lA.hat-lB.hat))/(sqrt((lambda_n.hat*2)/Shi_n.hat))
T2[z]=(sqrt(N)*(lA.hat^(2/3)-lB.hat^(2/3)))/((2/3)*sqrt((2*lambda_n.
hat^(1/3))/Shi_n.hat))
}
T0=sort(T1,decreasing=TRUE)
T01=sort(T2,decreasing=TRUE)
m=mean(propA)
sd=sd(propA)
crit_val_1=-8.4968      # ----> T0[M-(M*0.05)]=-8.4968(2,2)
crit_val_2=-7.94489359 # ----> T01[M-(M*0.05)]
test=0
for(t in 1:M)

```

```

{
  if(T1[t]<crit_val_1)
  {
    test=test+1
  }
  else
    test=test
}
test1=0
for(t in 1:M)
{
  if(T2[t]<crit_val_2)
  {
    test1=test1+1
  }
  else
    test1=test1
}
P=test/M      #power corresponding to 1st test statistic
P1=test1/M    #power corresponding to 2nd test statistic
#-----for calculating bias:
R.hat1=function(m)
{
  X_a=rpois(m,lambda=lambda_A)
  X_b=rpois(m,lambda=lambda_B)
  p1=0
  p2=0
  for(j in 1:m)
  {
    if(X_a[j]<X_b[j])
      p1=p1+1
    else if(X_a[j]==X_b[j])
      p2=p2+1
    else
    {
      p1=p1
      p2=p2
    }
  }
  Ra.hat1=(p1/m)+0.5*(p2/m)
  return(Ra.hat1)
}
bias=mean(propA)-R.hat1(5000)
mse=(sd(propA)^2)+(bias^2)
return(c(m,sd,bias,mse,crit_val_1,crit_val_2,P,P1))
}

```


The calculations for Power calculation and consistency are given as

```
myfunction(100,1000,2,3)

#power calculation for test statistic 1

p1=myfunction(100,1000,2,2.4)[7]
p2=myfunction(100,1000,2,2.8)[7]
p3=myfunction(100,1000,2,3.2)[7]
p4=myfunction(100,1000,2,3.6)[7]
p5=myfunction(100,1000,2,4)[7]
p6=myfunction(100,1000,2,4.4)[7]
p7=myfunction(100,1000,2,4.8)[7]
p8=myfunction(100,1000,2,5.2)[7]
p9=myfunction(100,1000,2,5.6)[7]
p10=myfunction(100,1000,2,6)[7]
p=c(p1,p2,p3,p4,p5,p6,p7,p8,p9,p10)

#power calculation for test statistic 2

p11=myfunction(100,1000,2,2.4)[8]
p12=myfunction(100,1000,2,2.8)[8]
p13=myfunction(100,1000,2,3.2)[8]
p14=myfunction(100,1000,2,3.6)[8]
p15=myfunction(100,1000,2,4)[8]
p16=myfunction(100,1000,2,4.4)[8]
p17=myfunction(100,1000,2,4.8)[8]
p18=myfunction(100,1000,2,5.2)[8]
p19=myfunction(100,1000,2,5.6)[8]
p20=myfunction(100,1000,2,6)[8]
p_1=c(p11,p12,p13,p14,p15,p16,p17,p18,p19,p20)
x=seq(2.4,6,0.4)

#same test statistic but different functions

plot(x,p,type="l",col="blue",lwd=2,main="Comparison of power of the
      test for lambda A=2 and different values of lambda B",xlab="Lambda B",
      ylab="Power",ylim=c(0,max(max(P1),max(p))))
lines(x,P1,col="red",lwd=2)
legend("topleft",col=c("red","blue"),legend=c("e^u/(1+e^u)","a^-u"),lty
      =1:1,lwd=2:2,bty="n")
grid()

#same function and dist but different test statistics

plot(x,p,type="l",col="green",lwd=2,main="Comparison of power for two
      seperate test statistics with lambdaA=2 and for different values of
      lambda B[a^-u]",xlab="Lambda B",ylab="Power",ylim=c(0,max(max(p_1),
      max(p))))
lines(x,p_1,col="purple",lwd=2)
legend("topleft",col=c("green","purple"),legend=c("test_stat_1","test_
      stat_2"),lty=1:1,lwd=2:2,bty="n")

#consistency for test stat 1

c1=myfunction(100,1000,2,4)[7]
c2=myfunction(150,1000,2,4)[7]
```

```

c3=myfunction(200,1000,2,4)[7]
c4=myfunction(250,1000,2,4)[7]
c5=myfunction(300,1000,2,4)[7]
c6=myfunction(350,1000,2,4)[7]
c7=myfunction(400,1000,2,4)[7]
C=c(c1,c2,c3,c4,c5,c6,c7)

#consistency for test stat 2

C1=myfunction(100,1000,2,4)[8]
C2=myfunction(150,1000,2,4)[8]
C3=myfunction(200,1000,2,4)[8]
C4=myfunction(250,1000,2,4)[8]
C5=myfunction(300,1000,2,4)[8]
C6=myfunction(350,1000,2,4)[8]
C7=myfunction(400,1000,2,4)[8]
C_1=c(C1,C2,C3,C4,C5,C6,C7)
s=seq(100,400,50)
plot(s,C,type="l",col="red",lwd=2,main="Checking consistency of the
      test based on two test statistics[for a^-u]",xlab="sample size",ylab
      ="Power",ylim=c(min(C),max(C)))
lines(s,C_1,col="green",lwd=2)
legend("topleft",col=c("red","green"),legend=c("test_stat_1","test_stat
      _2"),lty=1:1,lwd=2:2,bty="n")
grid()

```

• The rest of the codes are given through the google drive folder. power curve is calculated using different distributions as well as over different functions for a better comparison and valuable interpretation.

checkout the codes here — [Project Codes Google Drive](#) .

References

- [1] Wikipedia contributors, *Clinical Trials*, Wikipedia, The Free Encyclopedia. Available at: https://en.wikipedia.org/wiki/Clinical_trial
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- [3] Garzón-Orjuela, N., Garcia Pereira, A., Vornhagen, H., Stasiewicz, K., Parveen, S., Amin, D., Porwol, L., d'Aquin, M., Collins, C., Stanley, F., O'Callaghan, M., & Vellinga, A. (2024), *Design and architecture of the CARA infrastructure for visualising and benchmarking patient data from general practice*, BMJ Health & Care Informatics, 31(1), e101059. <https://doi.org/10.1136/bmjhci-2024-101059>
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